

- Concise Functional Requirements Document (FRD) for the "Application"***
- Introduction:**

The Application is designed to integrate customer and order data from CSV files into Delta tables, perform data cleansing and transformation, and create aggregated reports.

- Functional Requirements:**

1. **Data Ingestion Module**

- Read CSV data from specified volumes:
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`.
- Load data into Delta tables `customer` and `order`.

2. **Data Cleansing Module**

- Remove "Null"/Null and duplicate records from `customer` and `order` tables.

3. **Data Transformation and Loading Module**

- Create `ordersummary` table if it does not exist with the specified schema.
- Join `customer` and `order` data using "CustId" and load into SCD Type 2 table `ordersummary`.
- Update SCD Type 2 table `ordersummary` on changes in `customer` table.

4. **Aggregated Reporting Module**

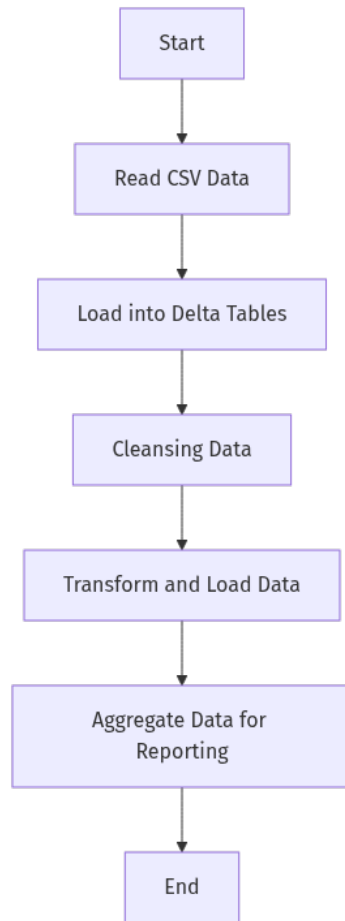
- Create `customeraggregatespend` table if it does not exist with the specified schema.
- Aggregate "TotalAmount" from `ordersummary` and load into `customeraggregatespend`.

- Detailed Functional Requirements:**

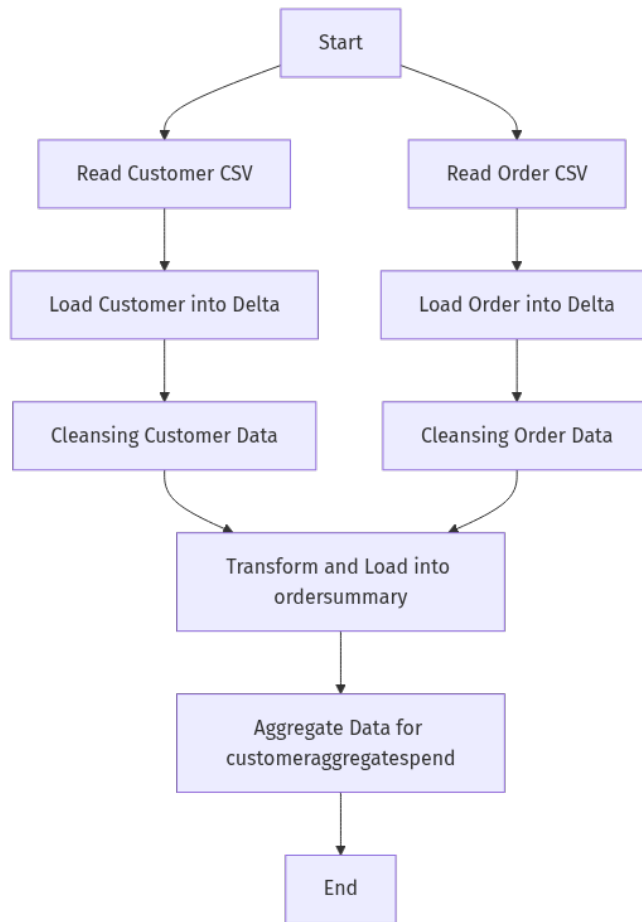
1. The system shall read CSV data from the specified volumes and load it into Delta tables `customer` and `order`.
2. The system shall remove "Null"/Null and duplicate records from both `customer` and `order` tables.
3. The system shall create the `ordersummary` table if it does not exist and load the joined data into an SCD Type 2 table.
4. The system shall update the SCD Type 2 table `ordersummary` whenever there is a change in the `customer` table.
5. The system shall create the `customeraggregatespend` table if it does not exist and load aggregated data into it.

- Diagrams:**

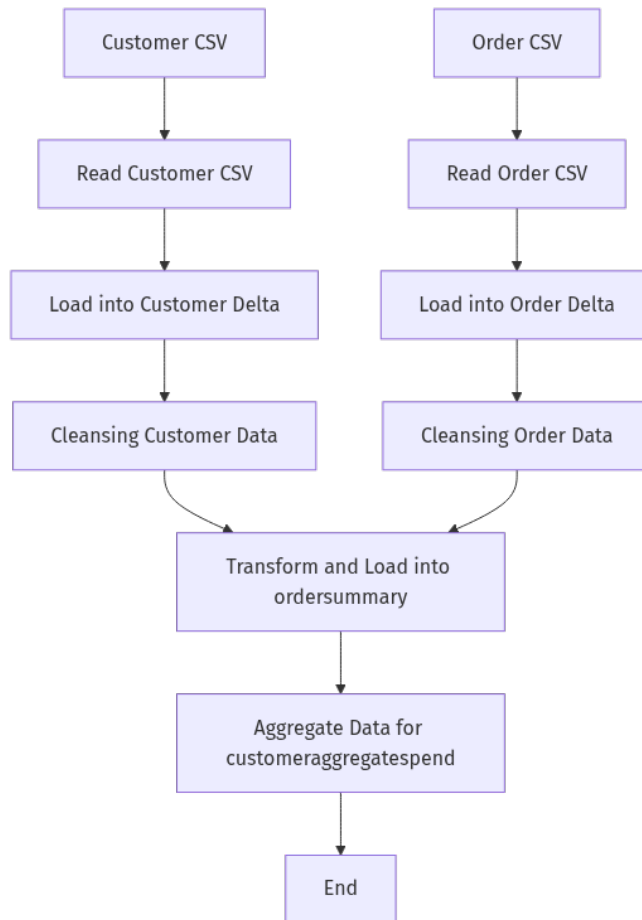
System Flow Diagram



Process Flow / Flowchart



Data Flow Diagram (DFD)



Sequence Diagram

[Diagram failed: Kroki render failed (HTTP 400): Error 400: SyntaxError: Parse error on line 2: ...aph TD;participant CSV_Customer as "Cus -----^ Expecting 'SEMI', 'NEWLINE', 'EOF', 'AMP', 'START_LINK', 'LINK', 'LINK_ID', got 'NODE_STRING' Error: Syntax error in graph at Worker.convert (file:///usr/local/kroki/src/worker.js:44:15) at async file:///usr/local/kroki/src/index.js:31:28]

High-Level Architecture Diagram

